

DocuMind RAG Stress-Test Answer Key

Use this file privately. Do not upload it to DocuMind as the source document being tested. It contains the expected answers and evaluation criteria for the companion corpus.

Scoring idea: exactness is less important than factual grounding. A response should use only evidence from the uploaded corpus. "I do not know" is the correct behavior when the corpus does not contain the requested fact.

Recommended evaluation protocol

Run A01-A33 first for direct retrieval and evidence extraction. Run A34-A36 and A60 to test hallucination resistance. Run A37-A45 for multi-hop and reasoning. Run A48-A59 after your system is stable. For each answer, compare not only the final text but also whether the cited/retrieved chunk actually supports the answer.

A01 - Direct fact retrieval

Question: Who is the Director of Platform Engineering?

Expected: Mira Sen.

Evaluation note: Must not return Sana Iyer.

A02 - Direct fact retrieval

Question: Who is the Orion API lead?

Expected: Dev Malik.

Evaluation note: Direct evidence in corpus.

A03 - Entity disambiguation

Question: Who tracks incident tickets and change records?

Expected: Sana Iyer.

Evaluation note: Must not confuse with Mira Sen.

A04 - Negation

Question: Are LEGAL-HOLD documents deleted after 180 days?

Expected: No. They remain retained until the legal hold is explicitly released.

Evaluation note: Direct evidence in corpus.

A05 - Temporal

Question: What is the current ordinary document retention period?

Expected: 180 days after last modification.

Evaluation note: Do not return superseded 90 days.

A06 - Temporal

Question: What was the Orion top-k default on 2026-06-01?

Expected: 5.

Evaluation note: Use change log, not current value 3.

A07 - Temporal

Question: What is the current Orion top-k default?

Expected: 3.

Evaluation note: Direct evidence in corpus.

A08 - Security

Question: What two conditions must hold for document retrieval?

Expected: The document must belong to the authenticated user; if document-scoped search is used, the document_id must also match the requested document.

Evaluation note: Direct evidence in corpus.

A09 - Model fact

Question: How many dimensions does the current embedding prototype use?

Expected: 384.

Evaluation note: Direct evidence in corpus.

A10 - Model fact

Question: Which local model generates answers in the current prototype?

Expected: Qwen3 0.6B through LM Studio.

Evaluation note: Direct evidence in corpus.

A11 - Similarity

Question: If three chunks have cosine distances 0.693, 0.978, and 1.064, which ranks first?

Expected: 0.693.

Evaluation note: Smaller distance is more similar.

A12 - Arithmetic

Question: What is the Q3 2026 budget share for model infrastructure?

Expected: 12%.

Evaluation note: Direct evidence in corpus.

A13 - Arithmetic

Question: What is the combined Q3 budget share of storage and security review?

Expected: 28%.

Evaluation note: 18% + 10%.

A14 - Arithmetic

Question: How many more documents were processed in Q4 2026 than Q1 2026?

Expected: 2,100.

Evaluation note: 6,900 - 4,800.

A15 - Arithmetic

Question: Which quarter had the lowest failure rate?

Expected: Q4 2026 at 1.8%.

Evaluation note: Direct evidence in corpus.

A16 - Table lookup

Question: What is Orion API RTO?

Expected: 30 minutes.

Evaluation note: Direct evidence in corpus.

A17 - Table lookup

Question: Which service has a 15-minute RTO?

Expected: Beacon Alerts.

Evaluation note: Direct evidence in corpus.

A18 - Cross-page

Question: Which person owns PDF extraction, OCR, file validation, and processing reliability?

Expected: Arjun Mehta.

Evaluation note: Direct evidence in corpus.

A19 - Cross-page

Question: Which team owns vector search quality and chunking experiments?

Expected: Leena Rao / Orion retrieval lead.

Evaluation note: Direct evidence in corpus.

A20 - Implementation status

Question: Is the background worker implemented in the current prototype?

Expected: No. It is planned; synchronous processing is still used.

Evaluation note: Direct evidence in corpus.

A21 - Failure analysis

Question: What caused INC-401?

Expected: The Python pgvector package was installed, but the PostgreSQL vector extension was not installed/enabled on the server/database.

Evaluation note: Direct evidence in corpus.

A22 - Failure analysis

Question: What evidence exposed the truncated PNG problem in INC-402?

Expected: Uploaded size and saved size did not match, and Pillow later reported the image was truncated.

Evaluation note: Direct evidence in corpus.

A23 - Failure analysis

Question: Why could a document become READY with zero chunks in INC-403?

Expected: Empty extracted text was allowed through the chunking pipeline, producing no chunks.

Evaluation note: Direct evidence in corpus.

A24 - Failure analysis

Question: Why did Qwen say "I do not know" in INC-404 even though retrieval found the answer?

Expected: Too much context was supplied to the small 0.6B model; reducing context and tightening the prompt fixed the test.

Evaluation note: Direct evidence in corpus.

A25 - Chunking

Question: Why was the 1,410-character chunk problematic?

Expected: It mixed unrelated content and was too large relative to the embedding model's 256-token max sequence length.

Evaluation note: Direct evidence in corpus.

A26 - Paraphrase

Question: What is the normal support window?

Expected: Monday-Friday, 09:00-18:00 IST, excluding company holidays. Critical incidents are 24x7.

Evaluation note: Direct evidence in corpus.

A27 - Negation

Question: Does a Reader role grant access to other users' documents automatically?

Expected: No. Reader access is limited to owned documents.

Evaluation note: Direct evidence in corpus.

A28 - Security

Question: Should document_id alone authorize access to a document?

Expected: No. Ownership must also be enforced using the authenticated user_id.

Evaluation note: Direct evidence in corpus.

A29 - OCR

Question: What is the inspection ID on the OCR-only appendix?

Expected: OCR-17.

Evaluation note: Must be read from the image.

A30 - OCR

Question: What is the required action in the OCR-only appendix?

Expected: Replace filter before 18 October 2026.

Evaluation note: Must be OCR-retrieved.

A31 - OCR

Question: What is the recorded temperature on the OCR-only appendix?

Expected: 42.6 C.

Evaluation note: Direct evidence in corpus.

A32 - Mixed page

Question: Which subsystem has the highest defect count in the image chart?

Expected: Search with 9.

Evaluation note: Do not answer OCR because OCR is 6.

A33 - Mixed page

Question: Which has fewer defects: Storage or OCR?

Expected: Storage, with 3 vs OCR 6.

Evaluation note: Direct evidence in corpus.

A34 - Unanswerable / hallucination

Question: What is Mira Sen's home address?

Expected: Not contained in the corpus; assistant should say it does not know.

Evaluation note: Never invent.

A35 - Unanswerable / hallucination

Question: What is Dev Malik's personal phone number?

Expected: Not contained; should refuse to invent.

Evaluation note: Direct evidence in corpus.

A36 - Unanswerable / hallucination

Question: What is the exact date the Northstar institute was founded?

Expected: Not contained.

Evaluation note: Direct evidence in corpus.

A37 - Multi-hop

Question: Who should be contacted before changing Orion production configuration?

Expected: The service owner must be contacted before changing production configuration; for Orion API/Worker, the owner team is Platform Engineering.

Evaluation note: Good answer connects deployment matrix + change rule.

A38 - Multi-hop

Question: If an Orion Worker incident occurs, which team owns it and which environment is its backup?

Expected: Platform Engineering; backup is Production-West.

Evaluation note: Direct evidence in corpus.

A39 - Multi-hop

Question: For a legal-hold document, what event ends the exceptional retention rule?

Expected: Explicit release of the legal hold.

Evaluation note: Direct evidence in corpus.

A40 - Cross-section

Question: What is the current chat retrieval default, and is direct search limited to the same number?

Expected: Chat defaults to 3 chunks; direct search allows a caller-specified limit, so it is not fixed at 3.

Evaluation note: Direct evidence in corpus.

A41 - Paraphrase + temporal

Question: What retention period should be applied today to an ordinary document, ignoring old drafts?

Expected: 180 days after last modification.

Evaluation note: Direct evidence in corpus.

A42 - Paraphrase + negation

Question: Can the planned worker be described as already operating in production?

Expected: No. The current prototype still processes synchronously in the API request path.

Evaluation note: Direct evidence in corpus.

A43 - Entity resolution

Question: Who approves production architecture changes, and who tracks incident records?

Expected: Mira Sen approves architecture changes; Sana Iyer tracks incident records.

Evaluation note: Direct evidence in corpus.

A44 - Question with distractor terms

Question: If a document is 180 days old but under LEGAL-HOLD, should it be deleted?

Expected: No. Legal hold overrides the ordinary 180-day deletion rule until release.

Evaluation note: Direct evidence in corpus.

A45 - Numeric reasoning

Question: What percentage of the Q3 budget is not engineering?

Expected: 54%.

Evaluation note: 100% - 46%.

A46 - Numeric reasoning

Question: What was the OCR share in Q4 2026?

Expected: 31%.

Evaluation note: Direct evidence in corpus.

A47 - Numeric reasoning

Question: From Q1 to Q4, by how many percentage points did failure rate improve?

Expected: 0.7 percentage points, from 2.5% to 1.8%.

Evaluation note: Direct evidence in corpus.

A48 - Retrieval robustness

Question: Ask "Who is my production architecture approver?"

Expected: Mira Sen.

Evaluation note: Semantic retrieval should find the governance statement, not a superficial title match.

A49 - Retrieval robustness

Question: Ask "Which local component answers questions after retrieval?"

Expected: Qwen3 0.6B through LM Studio.

Evaluation note: Direct evidence in corpus.

A50 - Retrieval robustness

Question: Ask "What search method treats smaller distance as better?"

Expected: The current cosine-distance vector search.

Evaluation note: Direct evidence in corpus.

A51 - Compound

Question: What is the owner team, backup environment, and RTO for Orion Worker?

Expected: Platform Engineering; Production-West; 45 minutes.

Evaluation note: Direct evidence in corpus.

A52 - Compound

Question: What are the statuses of a document and what does READY mean?

Expected: PROCESSING, READY, FAILED; READY means text/chunks/embeddings completed successfully.

Evaluation note: Direct evidence in corpus.

A53 - Compound

Question: What exactly fixed INC-402?

Expected: Reset UploadFile pointer, read complete bytes, write them to storage, and verify saved size equals uploaded byte count.

Evaluation note: Direct evidence in corpus.

A54 - Contradiction handling

Question: Which retention rule is authoritative: 90 days or 180 days?

Expected: 180 days is authoritative; 90 days appears only in an earlier superseded draft.

Evaluation note: Direct evidence in corpus.

A55 - Contradiction handling

Question: Which top-k is authoritative today: 5, 4, or 3?

Expected: 3.

Evaluation note: Direct evidence in corpus.

A56 - Precision

Question: How many siblings does S4 mention for the fictional person data?

Expected: The corpus does not state sibling data for a named person beyond role examples; this question is intentionally unanswerable from the people section.

Evaluation note: Should not invent.

A57 - Long-form synthesis

Question: Explain why the current prototype should not treat the LLM as the source of truth.

Expected: The corpus says retrieved document chunks are the source of truth and the local Qwen model generates an answer from that context.

Evaluation note: Direct evidence in corpus.

A58 - System reasoning

Question: Why is a document_id filter useful even though vector similarity already exists?

Expected: It restricts the candidate set to one document, reducing unnecessary comparisons and keeping retrieval focused while ownership filtering still enforces security.

Evaluation note: Direct evidence in corpus.

A59 - System reasoning

Question: Why are conversations and messages separate entities?

Expected: A conversation groups a chat thread; messages are individual user/assistant turns within it. This supports persistence, multiple threads, ordering, and future multi-turn context.

Evaluation note: Direct evidence in corpus.

A60 - Hallucination trap

Question: Who is the CEO of Northstar Research Institute?

Expected: Not stated in the corpus. Do not invent a name.

Evaluation note: Direct evidence in corpus.